# Resolution-Aware Lexicographic Tail-Regret Selection under Uncertain Normalisation Bounds

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**Abstract**

Selecting a recommendation from a multicriteria candidate set requires a declared preference rule and a treatment of uncertain normalisation bounds. Exact lexicographic minimax gives arbitrarily small worst-case differences absolute priority, hard to justify at finite decision resolution. We introduce ReLexTail, a preorder that first compares fixed-resolution categories of maximum and upper-tail active-range probe disappointments, then applies exact tail and ordered-profile refinements. The construction is complete and transitive, Pareto-compatible under monotone separating probes, and invariant to uniform replication of the probe multiset. A sufficient category-stability condition is distinguished from stability of the fully refined recommendation, and a sequential linear and mixed-integer characterisation covers continuous polyhedra. We then certify the decision rather than every coordinate. We prove that a recommendation can be certified throughout a bound box with no candidate retaining any category, and that keeping a probe's shared normalisation anchors inside a single quotient yields an enclosure contained in the independent one. Rational examples show both gaps are real: the all-candidate condition certifies nothing on an instance whose decision is certain over the whole box, and anchor slack alone decides whether another box is certified. On a benchmark of 192 candidate sets with splits locked before the run, the two changes raise the certified decision radius by 0.0330 and 0.0204 at a matched budget, and an exact oracle and an independent replay verifier falsified none of the emitted certificates. A separate 90-instance study with 500 matched perturbations at eight levels and 10,000 held-out preference vectors gives 19.30% point changes at resolution 0.02 against 20.88% for exact probe-lexicographic selection, with a paired interval including zero, and mean upper-tail regret 0.5411 against 0.5428. Grid shifts, ablations and a public building-simulation check expose boundary dependence. Decision quality against external preference populations and any calibration guarantee remain untested.

**Keywords:** multicriteria decision analysis; lexicographic selection; regret; conditional value-at-risk; normalisation uncertainty; robustness

## 1 Introduction

A multicriteria analysis often produces several efficient alternatives while a decision maker ultimately needs a recommendation. Selecting one alternative requires assumptions about acceptable trade-offs, even when those assumptions are expressed without an explicit criterion-weight vector. Weighted sums, reference-point methods and distance-based methods provide different ways of making these assumptions operational (Hwang & Yoon, 1981; Miettinen, 1999; Wierzbicki, 1980). Their recommendations also depend on the numerical scales on which criteria are compared. When normalisation bounds are estimated, robustness to those bounds is a distinct concern from uncertainty about preferences.

This paper considers a declared family of monotone questions about each alternative, called probes. A probe can assess one criterion, the mean across criteria, or the worst criterion. Its value is transformed into an active-range disappointment: zero denotes the best probe value available in the fixed candidate set and one the worst. A decision maker can inspect the named concerns

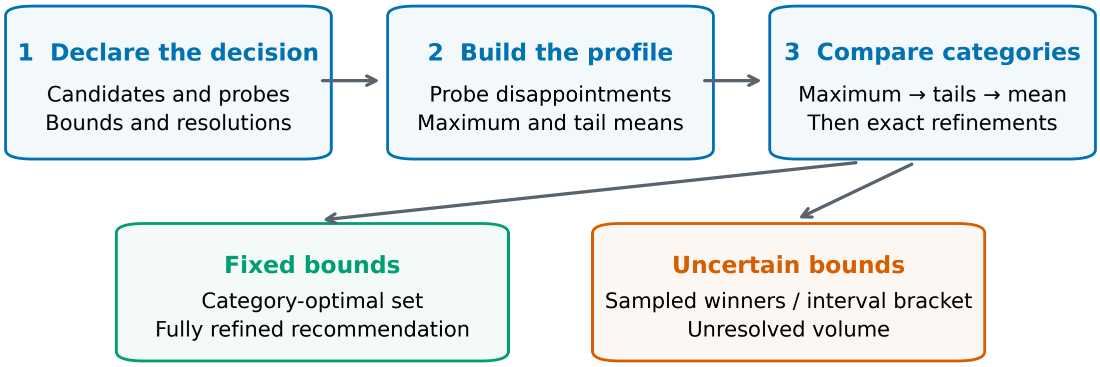


**Figure 1:** Overview of the decision procedure and graphical abstract. The declared model determines active-range disappointments. Category comparisons precede exact refinement. Under uncertain bounds, sampling and interval certification answer different questions and require different reports.

before accepting the recommendation. The probe family, its multiplicities, the candidate set and the normalisation convention remain substantive modelling choices.

Exact lexicographic minimax orders the sorted disappointments from worst to best. It gives any improvement in the largest disappointment priority over all improvements below it, regardless of the size of that leading difference. This is appropriate when priorities are strict. It can be difficult to justify when the decision model has a finite reporting resolution. Pairwise difference tolerances are an imperfect remedy because approximate equality need not be transitive. We instead use fixed resolution categories, followed by exact refinements, so that the comparison is a lexicographic order on a single well-defined vector.

The resulting rule, ReLexTail, first compares categories of the maximum and of upper-tail mean disappointments. Only when all category coordinates agree does it compare exact tail means and then the full sorted profile. Its contribution is this explicit hierarchy together with its decision-aiding interpretation, finite-set properties and treatment of uncertain normalisation bounds. The building blocks, including ordered aggregation, empirical conditional value-at-risk (CVaR) and lexicographic optimisation, are established tools. We do not claim their invention or a general superiority over other MCDA approaches.

Certifying a recommendation under uncertain bounds raises a question that is easy to answer too conservatively. The natural sufficient condition asks every candidate to keep every risk coordinate inside its resolution cell, and it fails whenever any coordinate touches a cell boundary – including a coordinate belonging to an alternative that is not in contention, a coordinate that is structurally invariant, and a coordinate occurring after the comparison has already been decided. None of those motions can change the recommendation. A second, independent source of conservatism sits in the arithmetic: the two normalisation anchors of a probe are shared by all candidates at a given bound vector, and widening the numerator and denominator of the disappointment ratio into separate intervals lets each take an extreme value the shared anchor could not have produced. Section 6 replaces both, proves that neither replacement weakens soundness, and measures what each is worth on a prespecified benchmark with locked splits.

The contribution is eightfold. First, we specify a fixed-cell tail-regret preorder with exact refinement, including boundary and tie conventions. Second, we establish finite-set existence, transitivity, conditional Pareto compatibility, replication invariance and a category-stability condition. Third, we distinguish sampled sensitivity from sound interval possible-winner elimination. Fourth, we give a sequential polyhedral formulation and its stage count. Fifth, we prove a decision-focused certification condition that requires no category to be retained, and exhibit a rational instance on which the all-candidate condition certifies nothing while the decision-focused condition certifies the entire uncertainty box without subdivision. Sixth, we prove that keeping the shared probe anchors inside a single quotient yields an enclosure contained in – and generically strictly inside – the independent one, and exhibit a rational instance where that difference decides whether the box is certified at all. Seventh, we test the quality–stability trade-off with matched perturbations, held-out preferences, paired uncertainty summaries, grid-origin sensitivity and ablations on 90 synthetic sets, plus a public building-simulation demonstration, and we test the certification claims on a separate locked benchmark with an exact rational oracle and an independent replay verifier. Eighth, we produce contrastive decision records and a release archive holding the data, scripts, current certification tests and documented reproduction commands, with known failures of the legacy test suite disclosed separately. These contributions combine established ordered-risk tools around finite decision resolution; they do not imply universal superiority or operational validation. The certification results say nothing about decision quality, external regret or calibration, which are not tested here. A ten-supplier illustration supplies an interpretable worked decision.

# 2 Related work

## 2.1 Multicriteria recommendation and ordered profiles

Multiple-criteria decision analysis offers several conceptually different routes from a set of efficient alternatives to a recommendation. Compensatory methods combine criterion evaluations through weighted scores, utility functions, distances or reference points, while noncompensatory and order-based methods emphasise the worst-performing coordinates or impose priorities between objectives (Hwang & Yoon, 1981; Miettinen, 1999; Wierzbicki, 1980). A recent retrospective traces this development from classical aggregation to preference disaggregation and robust ordinal regression (Greco, Słowiński, & Wallenius, 2025). That perspective matters here because ReLexTail is not offered as a universally preferable aggregation method: it implements a declared priority structure over a multiset of interpretable concerns.

Ordered weighted averaging assigns weights to ranked outcomes rather than to named criteria (Yager, 1988). Equitable optimisation models seek solutions that improve the least satisfactory components of an achievement vector (Kostreva & Ogryczak, 1999), and lexicographic minimax refines the comparison of ordered profiles by successively minimising their worst components (Ogryczak, 1997). Linear representations of ordered weighted objectives and of sums of the largest functions supply the computational basis for such models (Ogryczak & Śliwiński, 2003; Ogryczak & Tamir, 2003). Anytime lexicographic enumeration addresses a complementary problem: it generates lexicographically best nondominated solutions progressively (Foschini, Tsouros, Dilkina, & Guns, 2026), whereas ReLexTail assumes a fixed candidate set and defines a single preorder combining resolution categories, tail summaries and exact ordered-profile refinement. Its categories postpone sub-resolution differences, but the exact refinements still separate candidates within a category-optimal set, so the categories create no permanent indifference in the final preorder.

The upper-tail summaries used here are empirical CVaR-type functionals (Rockafellar & Uryasev, 2000). Their atoms, however, are declared probe disappointments, not realisations of a stochastic loss, and we write $\alpha$ for the upper-tail mass rather than a confidence level. *Tail regret* is accordingly a name for a decision index built from normalised disappointments; regret against an external preference population appears only in the evaluation of Section 7, in Equation (13). Keeping the two distributions separate is essential: a small tail of declared concerns need not imply a small tail of regret for an external decision maker. The same distinction separates the present setting from distributionally robust optimisation, which protects a decision against ambiguity in the probability law governing uncertain outcomes (Kuhn, Shafiee, & Wiesemann, 2025). ReLexTail treats a deterministic box of uncertain normalisation parameters and makes no distributional-robustness claim.

## 2.2 Preference uncertainty and normalisation uncertainty

Much of contemporary MCDA concerns incomplete, uncertain or interactively elicited preferences. Stochastic multicriteria acceptability analysis represents uncertainty over weights or evaluations through acceptability measures (Lahdelma, Hokkanen, & Salminen, 1998; Tervonen & Figueira, 2008). Robust ordinal regression derives necessary and possible preference relations from the set of value functions compatible with given preference information (Greco, Mousseau, & Słowiński, 2008). More recently, support procedures have been proposed for settings with unknown criterion weights (Więckowski & Sałabun, 2025), a Bayesian utility model with informative pairwise queries has been used to identify preferred Pareto solutions (Huber, Rojas Gonzalez, & Astudillo, 2025), and SMAA has been combined with FITradeoff to reduce the interaction burden of pairwise elicitation (Zhao, Balugani, Gamberini, & Lolli, 2026).

These methods vary preference parameters, criterion weights or probabilistic evaluations. ReLexTail addresses a different source: the preference rule, the probe family and the resolution grid are fixed, while the bounds used to normalise criterion values vary within a prescribed box. Its possible and necessary winner sets therefore borrow the quantifier structure of SMAA and robust ordinal regression while carrying different semantics. A possible ReLexTail winner is supported by at least one admissible normalisation vector, not by at least one compatible utility function or sampled weight vector.

Interactive elicitation is also shaped by behavioural and procedural choices. Anchoring can prevent users from reaching their most preferred solution in interactive multiobjective optimisation, and the effect changes with who guides the interaction (Halstead, López-Ibáñez, Farmer, & Warren, 2026). Such evidence reinforces the need to distinguish an explicitly declared deterministic rule from a claim of recovered latent preferences. ReLexTail provides the former. It does not claim that its cells or probes identify a decision maker's true utility function.

## 2.3 Robustness, sensitivity and certification

Sensitivity analysis measures how rankings change when weights, evaluations or method parameters are perturbed. Rank-stability and balance-point coefficients have been proposed for diagnosing the robustness of MCDM results (Paradowski, Wątróbski, & Sałabun, 2025), and the stability of several MCDM methods under perturbation has been compared in a logistics setting (Jangid, Kumar, Jahnvi, Dhanuk, & Sharma, 2025). From an optimisation perspective, relatively robust multicriteria decisions are defined through a worst-case performance ratio over admissible weights (Weber, 2026). Outranking has also been combined with Bayesian networks to represent uncertain evaluations (Cebesoy, Tuncer Şakar, & Yet, 2025).

These contributions show that "robustness" denotes no single mathematical property. It may mean empirical rank stability, worst-case performance over weights, probabilistic acceptability, or invariance of a selected alternative over a parameter region. ReLexTail adopts the last reading. Its certification question is whether the same unique recommendation is produced for *every* normalisation vector in a prescribed box. Random perturbation experiments give sensitivity evidence but cannot establish that universal statement, which is why Sections 5 and 6 report sampling and certification separately.

The interval component follows the standard principle that dependency loss enlarges enclosures when repeated occurrences of a shared variable are treated independently (Moore, Kearfott, & Cloud, 2009). The shared quantities here are the best and worst probe anchors that every candidate uses at a fixed bound vector. The contribution is therefore not a new principle of interval arithmetic but a decision-focused specialisation: the anchors are retained inside a common quotient, and certification stops at the first lexicographically decisive profile coordinate separating the nominal winner from each rival, instead of requiring every candidate and every risk coordinate to stay inside its original resolution cell.

## 2.4 Explanation and auditability

Explanation is now an explicit concern in MCDA. Contrastive explanations have been derived from fine-grained representations of the computations performed by weighted and hierarchical decision methods (Erwig & Kumar, 2025), and an explainable MCDA framework has been built on three-way decisions (Shi & Yao, 2025). ReLexTail shares the objective of making a recommendation inspectable, but its explanation object is specific to its preorder: for each rival it records the first coordinate at which the complete profile differs, the two values compared and, for categorical coordinates, the applicable resolution width. That record explains why one pairwise comparison is settled; independent replay of all pairwise records is what establishes optimality over the finite candidate set.

Auditability is not statistical calibration. A deterministic replay verifies that the published recommendation follows from the declared candidates, probes, bounds and numerical conventions.

**Table 1:** Structural positioning of neighbouring approaches. Entries describe the primary role of each family, not a quality ranking; implementations within a family can add further capabilities.

| Approach | Primary modelling object | Primary uncertainty | Typical output |
|---|---|---|---|
| Weighted score/distance | Compensatory score or distance | Weights/data if varied | Score or ranking |
| Leximax / LexPR | Ordered worst-to-best profile | None in the order | Exact optimal class |
| SMAA | Distribution over preferences/data | Preference/data sampling | Acceptability indices |
| Robust ordinal regression | Compatible preference models | Preference imprecision | Necessary/possible relations |
| DRO | Worst distribution in an ambiguity set | Probability-law ambiguity | Robust decision |
| ReLexTail | Fixed cells, tail summaries, exact profile | Bounded normalisation parameters | Category/exact winner sets |

It does not establish that the probes recover stakeholder preferences, that the selected candidate minimises an external population loss, or that a certificate carries a calibrated probability of correctness. Those remain separate empirical questions, and Section 8 records which of them this paper leaves open.

### 2.5 Positioning

Table 1 summarises the structural differences.

ReLexTail combines established components – active-range normalisation, ordered aggregation, empirical CVaR, fixed resolution cells, lexicographic refinement and interval enclosure – under a specific decision contract. Relative to exact lexicographic minimax, fixed cells postpone sub-resolution differences until broader tail categories have been compared. Relative to SMAA, Bayesian elicitation and robust ordinal regression, the preference rule is held fixed and the uncertainty is assigned to normalisation bounds. Relative to empirical robustness coefficients, the output is a deterministic box certificate rather than a sensitivity score. Relative to explainable MCDA, it supplies a method-specific first-difference record and an independent replay path. The closest methodological antecedent in our own earlier work is the integration of declared preferences into multiobjective scheduling (Bezoui, Olteanu, & Sevaux, 2023), which motivates interpretable preference modelling but supplies neither the present ordering nor its certificate.

The contribution is accordingly not the invention of CVaR, ordered aggregation, lexicographic optimisation or dependency-aware interval arithmetic. It is the construction and analysis of a resolution-aware preorder together with a decision-focused certificate that preserves the shared normalisation structure. We are not aware of prior work combining these elements under the same finite-set selection rule and certification contract. That is a scoped assessment of the literature we have examined, not a priority claim; the claim-to-evidence register in Section 8 records it as unestablished pending full-text verification of the closest work.

## 3 Representation and resolution-aware selection

### 3.1 Declared probes and active-range disappointment

Let $A = \{x_1, \ldots, x_N\}$ be a fixed, nonempty finite candidate set, and let $f_i(x)$, $i = 1, \ldots, m$, be criteria to minimise. For bounds $b = (z^\star, z^{\mathrm{nad}})$ with positive denominators, define

$$r_i(x; b) = \frac{f_i(x) - z_i^\star}{z_i^{\mathrm{nad}} - z_i^\star}. \tag{1}$$

Nominal bounds are the minimum and maximum of each criterion over $A$. Thus nominal values lie in $[0, 1]$. Perturbed or external bounds need not enclose the observed criteria, so perturbed values may lie outside that interval. We do not clip them. Constant criteria are excluded before normalisation and their exclusion is reported.

A probe $q$ is finite-valued and coordinatewise nondecreasing on a domain containing every reachable normalised vector. The declared family $\mathcal{Q}$ is a multiset. Its multiplicities act as weights in the empirical distribution of probe values. For each bound vector, define

$$q^\star(b) = \min_{x \in A} q(r(x;b)), \qquad q^w(b) = \max_{x \in A} q(r(x;b)), \qquad R_q(b) = q^w(b) - q^\star(b).$$

Retain probes with $R_q(b) > \varepsilon_q$, and set

$$D_q(x;b) = \frac{q(r(x;b)) - q^\star(b)}{R_q(b)} \in [0,1]. \tag{2}$$

The nominal computation uses $\varepsilon_q = 10^{-9}$. The ratio is evaluated without adding this threshold to the denominator. The threshold determines retention; it is not a smoothing constant. A run with no retained probes returns the entire candidate set as equivalent and cannot support a discriminating recommendation. All results below concern a nonempty retained family. Certification additionally requires a fixed retained family over its uncertainty box.

The canonical family contains single-criterion probes $q_i(r) = r_i$, the mean $q_\mu(r) = m^{-1} \sum_i r_i$, and the maximum $q_{\max}(r) = \max_i r_i$. There are at most $m+2$ retained probes. The mean may be constant on a particular candidate set and then contributes no information. We report the retained count rather than treating $m+2$ as an unconditional identity. Correlated criteria are not automatically merged, and no cluster probes are used in the computational study.

For a retained singleton, the two normalisations cancel exactly:

$$D_i(x;b) = \frac{f_i(x) - \min_{y \in A} f_i(y)}{\max_{y \in A} f_i(y) - \min_{y \in A} f_i(y)}. \tag{3}$$

Consequently, singleton disappointments are invariant to admissible bounds. Aggregate disappointments generally are not. Fixing external criterion bounds alone also does not provide independence from edits to $A$, because the probe anchors in Equation (2) still depend on that set. Full candidate-set independence would require fixing those anchors as well.

## 3.2 Tail means, categories and exact refinement

Write $D_{[1]}(x) \geq \cdots \geq D_{[K]}(x)$ for the retained disappointments sorted in decreasing order and $M(x) = D_{[1]}(x)$. For a tail fraction $\alpha \in (0,1]$, define

$$T_\alpha(x) = \min_{t \in \mathbb{R}} \left\{ t + \frac{1}{\alpha K} \sum_{q=1}^{K} [D_q(x) - t]_+ \right\}. \tag{4}$$

Here $\alpha$ denotes the mass of the upper tail, rather than its confidence level. For $h = \alpha K$, $k = \lfloor h \rfloor$ and $\theta = h - k$,

$$T_\alpha(x) = \frac{\sum_{j=1}^{k} D_{[j]}(x) + \theta D_{[k+1]}(x)}{h}.$$

The fractional term is omitted when $\theta = 0$. If $h < 1$, the expression equals $M(x)$. The fraction $\alpha = 1$ gives the overall mean. We use $T_{25} = T_{0.25}$, $T_{50} = T_{0.50}$ and $T_{100} = T_1$. Small retained families may make some tail coordinates redundant, without invalidating the rule.

Declare positive widths $\delta_M, \delta_{25}, \delta_{50}, \delta_{100}$. Categories are

$$B_\delta(z) = \lceil z/\delta \rceil, \qquad C_\ell(x) = B_{\delta_\ell}(R_\ell(x)), \quad R = (M, T_{25}, T_{50}, T_{100}). \tag{5}$$

Category zero contains zero. For $j \geq 1$, category $j$ is the interval $((j-1)\delta, j\delta]$. The boundary convention is part of the model and must be preserved in numerical replay.

For numerical replay, the implementation evaluates the quotient once in binary double precision, snaps it to the nearest integer only when the discrepancy is at most eight machine epsilons times $\max(1, |z/\delta|)$, and then applies the ceiling. This rule protects declared decimal boundaries such as 0.10/0.05 from representation noise; it is not a decision tolerance and does not move observations that are substantively near a boundary. Stable row identifiers settle any remaining full-profile tie.

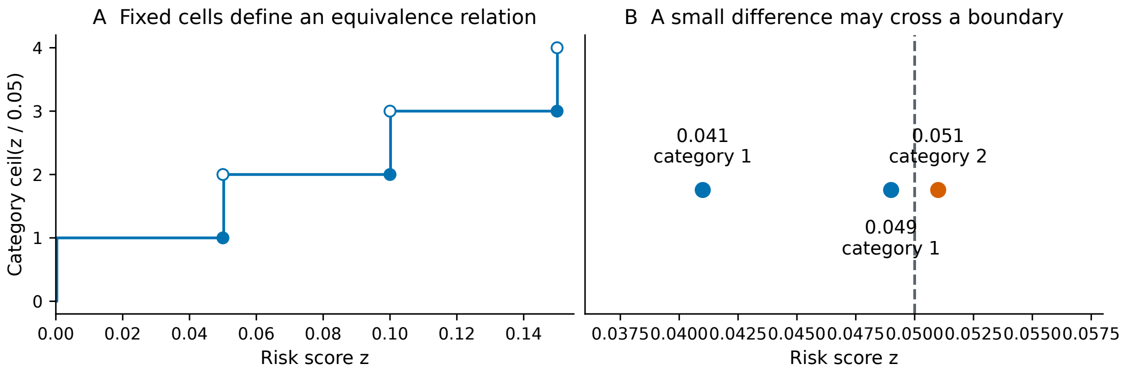


**Figure 2:** Fixed resolution cells and boundary sensitivity, illustrated at $\delta = 0.05$. Filled endpoints belong to the lower category and open endpoints to the upper category. Scores 0.049 and 0.051 are close but occupy different cells. This is a constructed explanation of the rule, not an empirical result.

## 3.3 Eliciting and auditing resolution widths

A resolution width is best interpreted as the smallest difference on an active-range concern that the decision maker is willing to give categorical priority. It is not estimated from the alternative that happens to win. One practical protocol is: (i) name the probes and show their best and worst attainable physical values; (ii) present controlled pairs that differ on one risk summary while other summaries are held as nearly equal as feasible; (iii) bracket the smallest difference that warrants crossing a priority level; and (iv) translate that bracket to the $[0, 1]$ active-range scale. Distinct widths may be elicited for maximum, upper-quarter, upper-half and mean concern. If the elicited answer is an interval, both endpoints and nearby values should be analysed rather than replaced by a favourable point estimate.

The reported analysis must then vary widths and the placement of cell boundaries. For a diagnostic origin fraction $\gamma \in [0, 1)$, we use

$$B_{\delta,\gamma}(z) = \left\lceil \frac{z - \gamma\delta}{\delta} \right\rceil,$$

with the same closed-upper-endpoint convention; because $z \geq 0$ and $\gamma < 1$, the first category remains labelled zero. The main specification is $\gamma = 0$. Shifts $\gamma \in \{0, .25, .5, .75\}$ test whether conclusions depend on an accidental boundary placement. This diagnostic does not substitute for eliciting a defensible physical resolution.

Define the full profile

$$\begin{aligned} \Psi(x) &= (C_M, C_{25}, C_{50}, C_{100},\ T_{25}, T_{50}, T_{100},\ M,\ D_{[2]}, \dots, D_{[K]})(x), \\ x \preceq_{\text{ReLexTail}} y &\iff \Psi(x) \leq_{\text{lex}} \Psi(y). \end{aligned} \tag{6}$$

Every category coordinate precedes every exact coordinate. Placing exact $M$ immediately after $C_M$ would restore strict maximum priority within each maximum category, defeating the intended comparison of broader tails. Exact refinement is instead postponed until the entire category prefix has been compared.

Let $W_{\text{cat}}$ minimise the four-coordinate category prefix and $W_{\text{exact}}$ minimise the full profile. Then $W_{\text{exact}} \subseteq W_{\text{cat}}$. Distinct alternatives are equivalent under the full preorder exactly when their sorted disappointment profiles coincide. When a single point is needed, we select the smallest candidate identifier in $W_{\text{exact}}$ and report this convention separately from the preference rule. The exact LexPR baseline minimises $D^{\downarrow}$ directly; deleting the category coordinates alone from Equation (6) does not produce LexPR, because the exact tail coordinates would still precede $M$.

Fixed cells resolve a logical issue, not all sensitivity. For scalar minimisation with $\delta = 0.05$, values 0, 0.04 and 0.08 give adjacent pairs within tolerance, whereas the first value is strictly preferable to the third. This demonstrates nontransitive approximate indifference, without establishing a strict preference cycle. Category equality is transitive. However, a very small change across a fixed

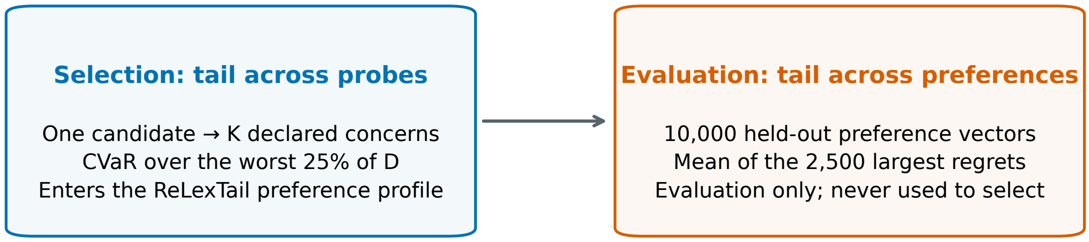


**Figure 3:** Two distinct uses of an upper tail. Selection aggregates the declared probes of a candidate. Evaluation aggregates range-normalised regret across held-out preference vectors. The latter weights are shared across methods and are never used to select a candidate.

boundary can alter a category, and exact refinement can change the point while the category-optimal set stays fixed. We therefore measure these outcomes separately.

### 3.4 Use of artificial intelligence

An AI assistant was used exclusively for minor grammatical and final textual corrections.

## 4 Properties and interpretation

**Theorem 1** (Existence and completeness)**.** *For every finite nonempty candidate set $A$, fixed finite retained probe family, and positive declared resolution vector, a ReLexTail-optimal alternative exists, and $\preceq_{\mathrm{ReLexTail}}$ is a complete preorder.*

*Proof.* $\Psi(x)$ is a finite-dimensional vector. Lexicographic order on these vectors is a complete preorder (not a total order, as distinct alternatives can have identical $\Psi$). The pullback through $\Psi$ is a complete preorder, which has at least one minimal element on a finite non-empty set. □

**Theorem 2** (Positional anonymity conditional on the declared multiset)**.** *For every permutation $\sigma$ of the probe coordinates under fixed probe multiplicities, $\Psi(\sigma D) = \Psi(D)$.*

*Proof.* $M$, empirical CVaR, and resolution categories depend only on symmetric quantities (order statistics). $D^{\downarrow}$ is permutation-invariant. Selective duplication changes the multiset and can change the result. □

**Proposition 1** (Strict profile monotonicity)**.** *If $D(x) \le D(y)$, then $x \preceq_{\mathrm{ReLexTail}} y$. If strictly less somewhere, $x \prec_{\mathrm{ReLexTail}} y$.*

*Proof.* Every coordinate of $\Psi$ is a nondecreasing functional of the coordinatewise input: empirical CVaR is monotone, $B_\delta$ is nondecreasing, and sorting is monotone, so $D(x) \le D(y)$ implies $\Psi(x) \le \Psi(y)$ coordinatewise, hence $\Psi(x) \le_{\mathrm{lex}} \Psi(y)$. If additionally $D_q(x) < D_q(y)$ for some $q$, then $T_{100}(x) < T_{100}(y)$, and no coordinate preceding $T_{100}$ in $\Psi$ can strictly favour $y$, so $\Psi(x) <_{\mathrm{lex}} \Psi(y)$. □

**Theorem 3** (Criterion-level Pareto compatibility)**.** *Fix a retained probe family $Q$. Assume every $q \in Q$ is coordinatewise nondecreasing and $Q$ is Pareto-separating on the reachable domain: whenever $r(x) \le r(y)$ and $r(x) \ne r(y)$, there exists $q \in Q$ such that $q(r(x)) < q(r(y))$. Then criterion-level Pareto dominance of $x$ over $y$ implies $x \prec_{\mathrm{ReLexTail}} y$.*

*Proof.* Since $x$ Pareto-dominates $y$ and the retained probes are nondecreasing, we have $D_q(x) \le D_q(y)$ for all $q$. By monotonicity of empirical CVaR, $T_\alpha(x) \le T_\alpha(y)$ for all $\alpha$, and since $B_\delta$ is nondecreasing, $C_\alpha(x) \le C_\alpha(y)$. Thus $\Psi(x) \le_{\mathrm{lex}} \Psi(y)$ and weak Pareto compatibility holds. For strictness, the Pareto-separating assumption gives a retained probe $q$ with $q(r(x)) < q(r(y))$. On that probe the active-range map $q(r(x)) \mapsto D_q(x)$ is a strictly increasing affine map (its slope is $1/R_q > 0$ because the probe is retained), so $D_q(x) < D_q(y)$, hence $D(x) \neq D(y)$. Since $D(x) \leq D(y)$ coordinatewise with at least one strict coordinate, $T_{100}(x) = \frac{1}{K}\sum_q D_q(x) < T_{100}(y)$. The coordinate $T_{100}$ precedes the terminal sorted block in $\Psi$ and no earlier coordinate can favour $y$, so $\Psi(x) <_{\text{lex}} \Psi(y)$ and $x \prec_{\text{ReLexTail}} y$. $\square$

**Theorem 4** (Uniform replication invariance)**.** *Let $rD$ denote the profile obtained by repeating every disappointment $r$ times. Under identical resolutions $\delta$ and tail fractions $\alpha$, $D \preceq_{\text{ReLexTail}} E \iff rD \preceq_{\text{ReLexTail}} rE$.*

*Proof.* The proof distinguishes the empirical distribution block from the terminal sorted-profile block. First, the empirical distribution of $rD$ is identical to $D$, meaning the empirical functionals remain unchanged: $M(rD) = M(D)$ and $T_\alpha(rD) = T_\alpha(D)$, so all resolution categories coincide. Second, if $D^{\downarrow}(x) <_{\text{lex}} E^{\downarrow}(y)$, the uniformly repeated sorted vectors preserve the first differing original coordinate and its direction, preserving the lexicographic comparison. $\square$

**Theorem 5** (Category stability)**.** *For any finite candidate set $A$ and fixed retained probe count $K$, empirical upper-tail CVaR and the maximum are 1-Lipschitz in the supremum norm over disappointment space. For any risk coordinate $R_\ell \in \{M, T_{25}, T_{50}, T_{100}\}$, define the distance to the nearest category boundary as $\mu_\ell(x) = \operatorname{dist}(R_\ell(x), \delta_\ell\mathbb{Z})$. Let $\mu_{\text{cell}} = \min_{x\in A,\ell}\mu_\ell(x)$. If a perturbation in the disappointment matrix satisfies $\|\widehat{D} - D\|_\infty = \eta < \mu_{\text{cell}}$, then every candidate remains in the same category on every category coordinate. Note that if perturbations occur in the bounds $b$ rather than directly in disappointments, a propagation bound $\|D(b) - D(\widehat{b})\|_\infty \leq L_B\|b - \widehat{b}\|$ must be applied.*

*Proof.* For the maximum $M$, the 1-Lipschitz property is immediate since $|M(D) - M(\widehat{D})| \leq \max_q |D_q - \widehat{D}_q| \leq \eta$. For empirical upper-tail CVaR, by definition of the perturbation bound, $D - \eta\mathbf{1} \leq \widehat{D} \leq D + \eta\mathbf{1}$. By monotonicity and translation equivariance of CVaR, $T_\alpha(D) - \eta \leq T_\alpha(\widehat{D}) \leq T_\alpha(D) + \eta$, yielding $|T_\alpha(D) - T_\alpha(\widehat{D})| \leq \eta$. Because the shift in every $R_\ell$ is at most $\eta < \mu_{\text{cell}}$, no value crosses a category boundary $\delta_\ell\mathbb{Z}$, preserving all categories. $\square$

**Remark (Exact-refinement fragility).** Stability of the fully refined ReLexTail point does not generally follow from a positive gap at the nominal decisive exact coordinate. If any preceding exact coordinate is tied and can change under perturbation, an arbitrarily small perturbation may reverse the lexicographic comparison. A positive full-winner radius requires additional structural invariance of the preceding ties or a direct interval-profile certificate.

The category condition is sufficient, not necessary. It can be vacuous when any risk coordinate lies exactly on a boundary, even if that coordinate is structurally invariant. For perturbations in physical bounds, a bound on the induced change in $D$ is additionally required. A distance measured only for one coordinate of the selected candidate cannot substitute for the minimum over all candidates and risk coordinates.

Nor are category-optimal sets generally nested when a common width is increased: cell boundaries move as well as merge, and later category coordinates can become decisive in a different way. Nesting follows only for specially aligned coarsenings at the level of each scalar partition, and even then it does not automatically transfer to the lexicographically optimal set. Resolution sensitivity must therefore be computed rather than inferred from set inclusion.

These properties describe the declared preorder and do not imply preference recovery, independence of the candidate set, or statistical optimality under an external loss distribution. In particular, equal replication of every probe preserves the empirical distribution, whereas duplicating only a subset does not.

# 5 Computation and uncertain bounds

## 5.1 Finite selection and comparison records

For finite $A$, evaluate each retained probe, compute its anchors, form $D$, sort each row, construct $\Psi$ and scan for the minimum. A second scan collects every exact and category tie. Computing the

**Table 2:** Uncertainty and tolerance sources in the reported workflow.

| Quantity | Role | Treatment |
|---|---|---|
| Resolution $\delta$ | Preference-model granularity | Declared fixed cells; varied in sensitivity analysis |
| Bound level $p$ | Epistemic normalisation uncertainty | Matched random draws or complete interval box |
| Retention $\varepsilon_q$ | Degenerate-probe guard | Fixed at $10^{-9}$; never added to a denominator |
| Machine/solver tolerance | Numerical implementation | Boundary snapping by eight machine epsilons for finite replay; solver tolerances must be reported separately |

canonical family costs $O(Nm)$ and sorting costs $O(NK \log K)$. Comparing profiles in a scan costs $O(NK)$. Thus selection costs $O(Nm \log m)$ for the canonical family. This complexity excludes generating candidates and Pareto filtering. It is an operation count, not a measured runtime guarantee.

Four numerical quantities have different meanings and are never substituted for one another. Table 2 records the distinction used in the algorithms and experiments.

Finite replay uses IEEE binary double precision for raw arithmetic, descending stable sorting, the declared boundary snapping rule in Section 3, and the smallest immutable row identifier only after equality of the full profile. The independent certification checks recompute category profiles from exact rational representations of outward-rounded interval endpoints. These conventions are recorded in the protocol file so that numerical tolerance cannot silently become a preference parameter.

For each rival $y$ of a unique winner $x^\star$, a contrastive record stores the first differing coordinate of $\Psi(x^\star)$ and $\Psi(y)$, the two values, the comparison direction and the resolution if that coordinate is categorical. Together with the full disappointment matrix, this permits independent comparison replay. A labelled disappointment plot alone explains concerns but does not prove optimality. Recomputing probe values additionally requires the criterion matrix, bounds, formulas and retained identifiers. Recreating the candidates requires the upstream model and seeds. These levels of replay should not be conflated.

## 5.2 Sampling and certified enclosures

For a compact admissible box $\mathcal{B}$, let $W(b)$ be the full exact optimal class. Define

$$S^{\mathrm{pos}} = \bigcup_{b \in \mathcal{B}} W(b), \qquad S^{\mathrm{nec}} = \bigcap_{b \in \mathcal{B}} W(b). \tag{7}$$

A sampled union is an inner approximation of the possible set. A sampled intersection is an outer approximation of the necessary set, not a certificate of necessary optimality. If a candidate wins on a region of sampling probability at least $a > 0$, then $B$ independent draws miss it with probability at most $(1-a)^B$. By a union bound, $B \geq \lceil \log(N/\epsilon)/[-\log(1-a)] \rceil$ detects all such candidates with probability at least $1-\epsilon$. It gives no guarantee for regions of zero or smaller probability.

Our experiments perturb each bound by a fraction of its nominal criterion range. If $I_i = \min_A f_i$, $N_i = \max_A f_i$ and $s_i = N_i - I_i > 0$, define

$$\mathcal{B}(p) = \prod_i [I_i - ps_i, I_i + ps_i] \times [N_i - ps_i, N_i + ps_i], \qquad 0 \leq p < \tfrac{1}{2}. \tag{8}$$

Denominators are bounded below by $(1-2p)s_i > 0$. This convention is invariant to a change of criterion origin. Frequency estimates use independent uniform draws for each bound coordinate. Certification concerns the complete box and does not depend on this sampling distribution. The supplier sampling illustration also includes $p = 0.5$, for which sampled denominators were checked positive; that endpoint is not used for certification because the closed box can contain zero denominators.

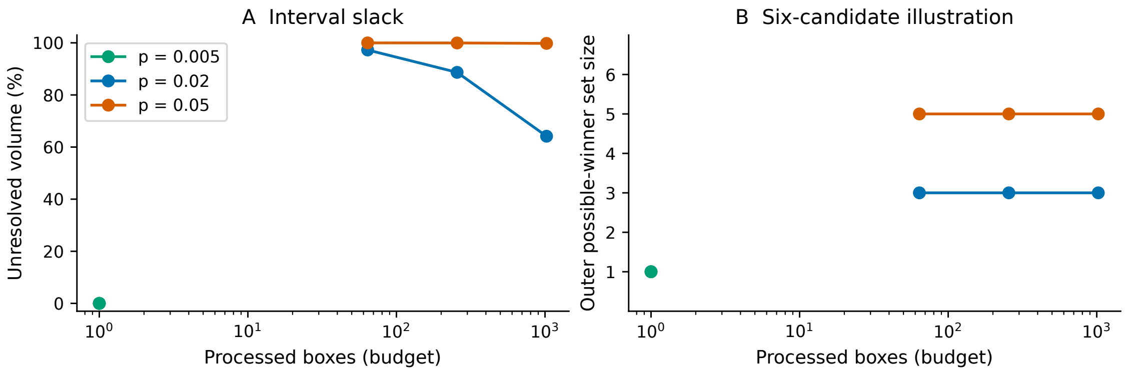


**Figure 4:** New interval illustration for six candidates and three criteria, with ReLexTail resolution 0.02. Each curve uses the same criterion matrix and range-relative uncertainty box. The horizontal axis reports processed boxes, which may be fewer than the budget after early termination. The narrowest box resolves immediately; larger boxes can retain substantial unresolved volume despite increased effort.

Consider a sub-box $\beta$. Assume every retained probe has active range uniformly greater than $\varepsilon_q$ throughout $\beta$. Singleton retention can be checked using

$$\inf_{b\in\beta} R_{q_i}(b) = \frac{s_i}{\sup_{b\in\beta}(z_i^{\mathrm{nad}} - z_i^{\star})}.$$

A nominal check alone is insufficient. For aggregates, bound $q(r(x; b))$ using coordinatewise enclosures and monotonicity, then enclose the anchors and the ratio in Equation (2). The extrema of each normalised criterion over its two-dimensional rectangle occur at corners when denominators remain positive. Dependency between the ratio terms may produce substantial slack. If uniform retention cannot be established, the box remains unresolved and is subdivided or returned with its inherited candidates.

**Proposition 2** (Sound elimination)**.** *Suppose the retained multiset is fixed and* $D^L(x;\beta) \le D(x;b) \le D^U(x;\beta)$ *for all* $b \in \beta$*. If*

$$\Psi(D^U(x;\beta)) <_{\mathrm{lex}} \Psi(D^L(y;\beta)),$$

*then* $x \prec_{\mathrm{ReLexTail}} y$ *throughout* $\beta$*.*

*Proof.* Maximum, tail means, category maps and order statistics are coordinatewise nondecreasing. Hence $\Psi(D(x;b)) \le \Psi(D^U(x;\beta))$ and $\Psi(D^L(y;\beta)) \le \Psi(D(y;b))$ componentwise. Combining these inequalities with the strict lexicographic inequality proves the result. □

The branch-and-bound procedure eliminates a candidate only when the displayed test succeeds. A box with one survivor contributes that winner to both inner and outer possible sets. Unresolved terminal boxes contribute all survivors to the outer set. Crucially, these contributions are combined by union over leaves: elimination on one box cannot remove a candidate that wins on another. A strict processed-box budget preserves all unprocessed leaves. This produces

$$\mathrm{In} \subseteq S^{\mathrm{pos}} \subseteq \mathrm{Out}$$

with an explicitly recorded unresolved volume fraction. The new illustration uses outward-rounded enclosures and exact rational tail means and category comparisons of the outward-rounded binary endpoints. A floating-point sampled winner is a diagnostic witness; it is not by itself an exact-real optimality certificate.

Persistent ties and winners confined to boundaries can prevent finite resolution. An inner set based only on sole-winner boxes can omit such winners even when they belong to the possible set. No general finite-budget exact recovery is claimed. If every terminal box is resolved to the same sole winner, necessary optimality is certified. A positive exact-refinement margin at one nominal point alone does not establish this result.

The illustration uses the first six candidates of the saved concave instance with three criteria and replicate zero. At each $p \in \{0.005, 0.02, 0.05\}$, budgets are 64, 256 and 1,024 processed boxes. We independently sampled 256 bound vectors per level and checked that their point winners belong to every reported outer set. Outer sets were also checked to be nested as the budget increased. These are consistency checks on one instance, not a proof of soundness or a broad scalability study. At $p = 0.005$ the root box is resolved. At $p = 0.05$, more than 99% of the volume remains unresolved after 1,024 boxes. Reporting this limitation is essential to interpreting the bracket.

### 5.3 A direct formulation on a continuous polyhedron

Let $\mathcal{X}$ be a nonempty bounded continuous polyhedron, with affine criteria and nondegenerate canonical probes. Compute criterion bounds and probe anchors over $\mathcal{X}$ and then hold them fixed. Singleton and mean disappointments are affine. Maximum-probe disappointment is convex piecewise linear. Its epigraph has a linear representation. Since every subsequent objective and bound is monotone in disappointments, these epigraphs can represent the required upper bounds without introducing artificial improvements.

For a risk coordinate $R_\ell$, use an integer variable $z_\ell \geq 0$ and constraint $R_\ell(x) \leq \delta_\ell z_\ell$. Minimising $z_\ell$ gives the least feasible category. Fix its optimum and continue to the next category. At an optimal category, a feasible point cannot have a lower actual category, because that would contradict the previous minimisation. This is why upper category bounds suffice despite the open lower cell endpoints.

For $T_\alpha$, introduce $t_\alpha \in \mathbb{R}$ and $u_{\alpha q} \geq 0$, with

$$u_{\alpha q} \geq D_q(x) - t_\alpha, \qquad t_\alpha + \frac{1}{\alpha K} \sum_q u_{\alpha q} \leq \delta_\alpha z_\alpha. \tag{9}$$

The maximum uses an epigraph $v \geq D_q(x)$ for every $q$. After the four category stages, minimise $T_{25}, T_{50}, T_{100}, M$ in sequence, retaining all earlier optima. For the terminal refinement, minimise the sums of the largest $p$ disappointments, $p = 2, \ldots, K$:

$$S_p(D) = \min_{t_p \in \mathbb{R},\ u_{pq} \geq 0} \left\{ p\, t_p + \sum_{q=1}^{K} u_{pq} : u_{pq} \geq D_q(x) - t_p \quad \forall q \right\}. \tag{10}$$

Once preceding ordered coordinates are fixed, minimising $S_p$ minimises the next coordinate. These formulations follow standard ordered-optimisation identities (Ogryczak & Śliwiński, 2003; Ogryczak & Tamir, 2003).

**Proposition 3** (Sequential formulation)**.** *For the stated continuous-polyhedral setting with $K = m+2$ nondegenerate canonical probes, ReLexTail is characterised by a finite sequence of LPs and four integer category stages. With no reuse of anchor computations and no redundancy elimination, it uses $6m + 8$ LP calls and four MILP calls.*

*Proof.* Criterion bounds require $2m$ LPs. The $m + 1$ affine probes require $2(m + 1)$ anchor LPs. Minimising the grand maximum requires one epigraph LP and maximising it requires $m$ affine maximisations. Preprocessing therefore uses $5m + 3$ LPs. Four exact risk refinements and $K - 1 = m+1$ ordered-sum refinements add $m+5$ LPs. The four category stages introduce bounded integer variables. Sequentially fixing optimal categories leaves closed risk sublevel sets, and the later continuous objectives attain their minima on compact feasible sets. The sequence therefore realises the specified lexicographic optimum. □

If the original feasible set includes integer variables, preprocessing and refinements can themselves require MILPs. If probes are dropped or redundant stages are removed, the count changes. Solver tolerances also change the operational optimisation problem and must be reported separately from resolution widths. Finally, anchors computed over a polyhedron generally differ from anchors over a finite sampled shortlist. Agreement between the resulting recommendations is not guaranteed. The present submission does not claim an independently validated end-to-end continuous MILP implementation.

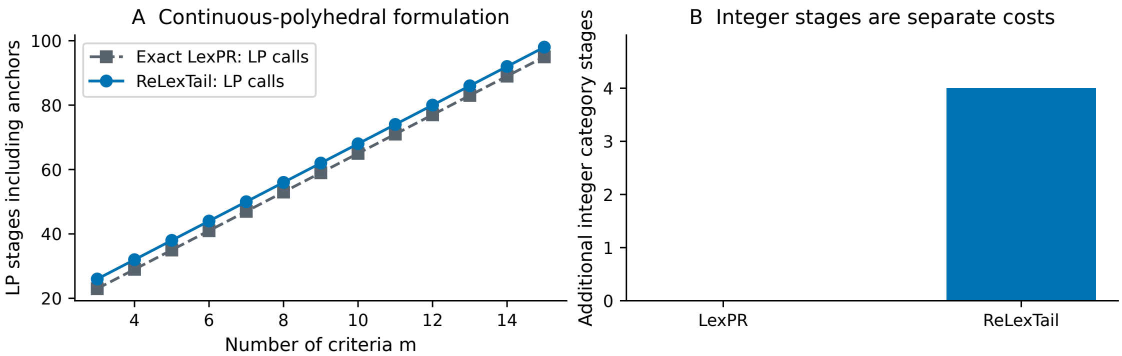


**Figure 5:** Theoretical solver-stage counts under the declared no-reuse continuous-polyhedral formulation. ReLexTail uses $6m + 8$ LP calls and four integer category stages; exact LexPR uses $6m + 5$ LP calls. These are formulation counts, not measured runtimes or evidence that MILP cost is constant.

# 6 Decision-focused certification

Section 5 certifies with a condition that asks more than the decision needs. Theorem 5 requires *every* candidate to keep *every* risk coordinate inside its cell, and Proposition 2 is applied through enclosures in which the shared normalisation anchors are widened independently for each candidate. This section replaces both with conditions matched to the question actually asked – does the recommendation survive? – proves that neither replacement weakens soundness, and measures what the change buys. The declared rule is untouched: on fixed inputs the engine of this section and the engine of Section 5 select the same alternative. What differs is what can be proved about that alternative and at what cost.

## 6.1 The decision region and its radius

Fix $A$, a retained multiset $\mathcal{Q}$, a resolution vector $\delta$ and the grid convention of Section 3. For a nominal winner $x_0$ that is unique, the *decision region* is

$$\mathcal{R}_{x_0} = \big\{b : \Psi(x_0; b) <_{\text{lex}} \Psi(y; b) \text{ for every } y \neq x_0\big\}, \tag{11}$$

and for the nested boxes $\mathcal{B}(p)$ of Equation (8) the *decision radius* is $\rho^\star(x_0) = \sup\{p : \mathcal{B}(p) \subseteq \mathcal{R}_{x_0}\}$. The computational object is a certified inner region for $\mathcal{R}_{x_0}$, reported together with the leaves that were not resolved. Three conventions matter and are not negotiable in what follows. A qualifying set that turns out to be empty is reported as “no unique-winner certificate”, never converted into positive stability. A tied nominal optimum requires a separately defined set-preservation contract and is recorded and skipped rather than resolved by tie-breaking. And the procedure is one-sided: failure to certify at level $p$ means only that the budget did not suffice, so an upper bound on $\rho^\star$ can come only from a verified counterexample, never from a failed search.

## 6.2 Certifying the decision, not every coordinate

**Proposition 4** (Decision-focused soundness)**.** *Let $\beta$ be a box on which the retained multiset is fixed, and let $D^L(\cdot;\beta) \le D(\cdot;b) \le D^U(\cdot;\beta)$ coordinatewise for every $b \in \beta$. If*

$$\Psi\big(D^U(x_0;\beta)\big) <_{\text{lex}} \Psi\big(D^L(y;\beta)\big) \qquad \textit{for every } y \neq x_0,$$

*then $\beta \subseteq \mathcal{R}_{x_0}$. No candidate is required to retain any category, and no coordinate of $\Psi$ beyond the first differing one is examined.*

*Proof.* Every coordinate of $\Psi$ is a nondecreasing functional of $D$ coordinatewise: order statistics, the maximum, the upper-tail means of Equation (4) and the category map $B_\delta$ of Equation (5) all are. Hence for $b \in \beta$, $\Psi(D(x_0;b)) \le \Psi(D^U(x_0;\beta))$ and $\Psi(D^L(y;\beta)) \le \Psi(D(y;b))$ componentwise, and componentwise inequality implies $\le_{\text{lex}}$. The lexicographic order is total and transitive, so $\Psi(x_0;b) \le_{\text{lex}} \Psi(D^U(x_0;\beta)) <_{\text{lex}} \Psi(D^L(y;\beta)) \le_{\text{lex}} \Psi(y;b)$ for every $y \neq x_0$ and every $b \in \beta$. $\square$

The chain uses no retention of categories anywhere. Category retention is therefore sufficient but not necessary, and the gap is not a technicality.

**Proposition 5** (The global condition can be vacuous while the decision is certain)**.** *There exist a candidate set with rational entries, a resolution vector and a level $p > 0$ such that every candidate changes at least one categorical coordinate between the lower and the upper enclosure over $\mathcal{B}(p)$ – so $\mu_{\text{cell}}$ of Theorem 5 certifies nothing – while the test of Proposition 4 certifies the unique winner against every rival on the whole of $\mathcal{B}(p)$ without any subdivision.*

*Proof.* By exhibition. Example E-A of the reproducibility archive is a $5 \times 4$ matrix whose entries are integer multiples of $1/64$, hence exact in binary double precision. At $p = 0.01$ all 5 candidates leave at least one cell, so the all-candidate condition refuses the root box; the pairwise test of Proposition 4 succeeds against all four rivals at that same root box. Both facts are recomputed by `certrelex.examples.check_boundary`. □

The mechanism is the one anticipated in Section 4: a coordinate may be free to move because it belongs to an alternative that is not in contention, because it sits on a boundary that is structurally invariant, or because it occurs after a comparison that is already decided. None of those motions can change the recommendation, and a condition that forbids them measures the wrong thing.

## 6.3 Preserving the shared normalisation anchors

Write $v_x(b) = q(r(x; b))$ for a retained aggregate probe. All candidates share the same two anchors $q^\star(b)$ and $q^w(b)$ at a given $b$, and Equation (2) reads $D_q(x; b) = g\big(v_x(b), q^\star(b), q^w(b)\big)$ with

$$g(v, a, c) = \frac{v - a}{c - a}. \tag{12}$$

Widening $v - a$ and $c - a$ into two independent intervals and dividing them lets the numerator take an extreme value against a denominator that the same $a$ could not have produced. Keeping $a$ inside a single expression removes that.

**Proposition 6** (Shared-anchor enclosure)**.** *Let $v \in [v^L, v^U]$, $a \in [a^L, a^U]$, $c \in [c^L, c^U]$ with $c^L > a^U$. Then $g$ is strictly increasing in $v$ and, for each fixed $v$, monotone in $a$ and in $c$ separately, with*

$$\frac{\partial g}{\partial c} = -\frac{v - a}{(c - a)^2}, \qquad \frac{\partial g}{\partial a} = \frac{v - c}{(c - a)^2},$$

*each of constant sign in the variable being varied. Consequently the range of $g$ over the product box is attained at the four corners of $[a^L, a^U] \times [c^L, c^U]$, evaluated at $v = v^L$ for the minimum and $v = v^U$ for the maximum. The resulting interval is contained in the independent quotient $[v^L - a^U,\ v^U - a^L]/[c^L - a^U,\ c^U - a^L]$, and the containment is strict whenever $a^L < a^U$ and $v^L \geq a^U$.*

*Proof.* The two derivatives are as displayed. For fixed $v$ the sign of $\partial g/\partial c$ is that of $-(v - a)$, which does not depend on $c$; the sign of $\partial g/\partial a$ is that of $v - c$, which does not depend on $a$. A function that is monotone in each variable separately on a rectangle attains its extrema at the rectangle's corners: minimise first over $c$ at an endpoint, then over $a$ at an endpoint. Monotonicity in $v$ places the minimum at $v^L$ and the maximum at $v^U$. For containment, note that each corner value is one of the four quotients $(v - a)/(c - a)$ with $a \in \{a^L, a^U\}$ and $c \in \{c^L, c^U\}$, and every such quotient lies in the independent interval because $v - a$ lies in $[v^L - a^U, v^U - a^L]$ and $c - a$ in $[c^L - a^U, c^U - a^L]$. If $v^L \geq a^U$ the corner minimum is $(v^L - a^U)/(c^U - a^U)$, whereas the independent lower endpoint is $(v^L - a^U)/(c^U - a^L)$; the denominators differ by $a^U - a^L > 0$ with a nonnegative numerator, so the inequality is strict, and symmetrically at the upper end. □

Two further identities are used and are not claimed as new. Singleton disappointments are bound-invariant by Equation (3), so $m$ of the at most $m + 2$ canonical coordinates carry no width however large the box is; the verifier of Section 6.4 rechecks that invariance at the box corners rather than assuming it. And branching is directed at the criteria whose normalised enclosure is widest among the candidates still involved in an unresolved comparison, because those are the only bound coordinates that can still move the decision; this changes where effort goes and not what is provable.

**Table 3:** Ablation ladder. Every arm certifies the same rule on the same inputs with the same budget of 300 processed boxes.

| Arm | Sufficient condition | Quotient | Branching |
|---|---|---|---|
| A0 | all-candidate retention | independent | widest coordinate |
| A1 | decision-focused | independent | widest coordinate |
| A2 | decision-focused | shared anchors | widest coordinate |
| A3 | decision-focused | shared anchors | decision-relevant |

**Proposition 7** (Cancellation-induced slack is not asymptotic)**.** *There exist a rational candidate set and a level $p > 0$ at which the enclosure of Proposition 6 certifies the unique winner on the whole of $\mathcal{B}(p)$ in a single box, while the independent quotient – identical in every other respect – leaves part of the box uncertified after a budget three orders of magnitude larger.*

*Proof.* By exhibition. Example E-B is a $4 \times 3$ matrix over the denominator 64. At $p = 0.05$ the joint enclosure resolves the root box in 1 box; the independent enclosure has certified only 33.0% of the volume after 2,048 processed boxes. The two runs differ only in Equation (12). Recomputed by `certrelex.examples.check_slack`. □

## 6.4 Anytime brackets, refusal and independent replay

Certified leaves are never revisited and unresolved leaves always retain their volume, so an interrupted run returns a valid certified region together with a valid bound on the uncertified volume, and continuing can only move volume from the second into the first. A box is *refused* outright when a normalising denominator or a probe active range is not certifiably positive on it; refusal leaves the box unresolved and never becomes positive stability.

Because the search is one-sided, an upper bound on $\rho^\star$ must come from elsewhere. We obtain it from an exact rational oracle that evaluates the declared rule at a single bound vector with no floating-point step: a $b$ at which the oracle names a different unique winner is a verified counterexample and bounds $\rho^\star$ from above. Example E-C is a $4 \times 3$ rational instance whose nominal winner is candidate 1; inside $\mathcal{B}(0.05)$ there is a bound vector at which the oracle names candidate 2 uniquely, while the engine certifies $\rho^\star \geq 0.0220$. The pair $[0.0220,\ 0.05]$ is a genuine two-sided bracket, which is what an honest anytime certificate should return in place of a single number.

Every certificate is replayed by a verifier written to be simpler than the search and independent of it: it imports nothing from the interval or enclosure modules, works entirely in exact rational arithmetic, and knows nothing about gates, budgets or branching. It is handed a leaf and a claimed winner and recomputes the enclosure and the comparison. A leaf both routes accept has been certified twice, by different arithmetic. The full numerical contract – reference rule, operational selector, certification arithmetic and verifier, and what each does *not* claim – is recorded in the archive.

## 6.5 Fixed-budget certification experiments

The study is a fixed-rule comparison: the same declared rule, probes, uncertainty box, arithmetic contract, box budget and hardware in every arm, so that any difference is attributable to the certification machinery and not to a different recommendation. Instances come from six development families and two locked stress families (near-tied margins; correlated heavy-tailed measurement noise) at $m \in \{3, 6, 10\}$ with eight replicates each, 192 candidate sets in all, certified on a shortlist of 10 alternatives. The split is a deterministic function of the replicate index and was fixed in source before any run; the 90 instances of the earlier sections remain the development baseline and are not used as confirmatory evidence. Resolution is $\delta = 0.02$ throughout and the focal level is $p = 0.02$. Each arm adds exactly one component to the one above:

The primary endpoint is the certified lower bound $\rho^{\mathrm{lo}}$ on the decision radius, obtained by bisection at the matched budget; it is continuous, paired within instance, and one value per arm. The two prespecified contrasts are P1 = A1 − A0, isolating the decision focus, and P2 = A2 − A1, isolating dependency preservation, both on the locked test split with Holm correction inside that

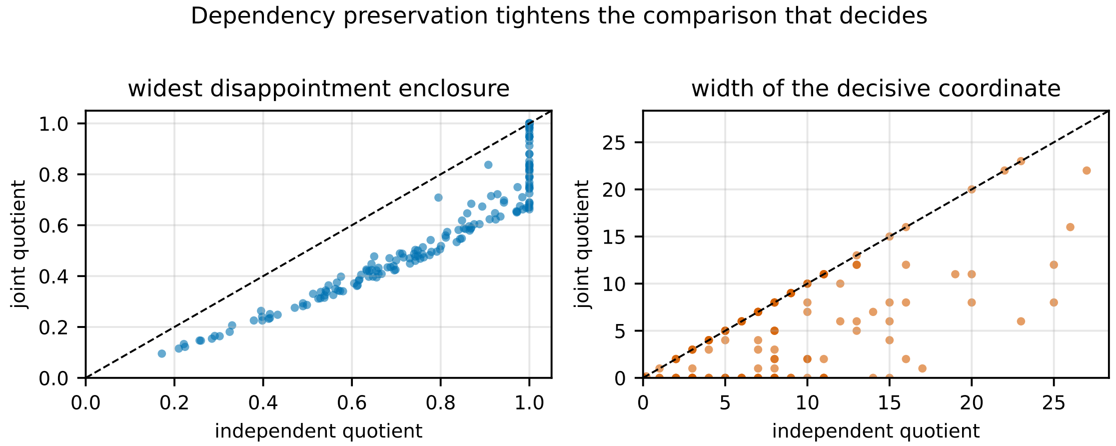


**Figure 6:** Dependency preservation tightens the comparison that decides. Each point is one instance at the root box $\mathcal{B}(0.02)$; the dashed line is equality. Left: widest disappointment enclosure. Right: enclosure width of the profile coordinate at which the nominal winner and its closest rival first differ. Points below the diagonal are instances where keeping the shared anchors inside one quotient narrows the enclosure.

family. Every other contrast is labelled exploratory. The independent unit is the instance. Because the study was not preregistered with a third party, $p$-values are descriptive.

**Correctness (E0).** Across 192 instances, 11,072 bound vectors were drawn from inside certified leaves and evaluated by the exact rational oracle, and 5,029 certified leaves were replayed by the independent verifier. There were 0 falsified certificates and 0 replay failures. Testing supplements Propositions 4 and 6; it does not substitute for them, and a zero count is evidence of the absence of the errors searched for, not of soundness in general.

**Mechanism (E1).** At the root box the widest disappointment enclosure averages 0.774 under the independent quotient and 0.582 under the shared-anchor quotient. The coordinate that decides the nominal comparison is categorical on 176 of the 178 instances with a resolvable root box; on those, the median enclosure width of that coordinate falls from 5.0 categories to 0.5. The remaining instances are decided on a tail mean and are excluded from that statistic because the two are not in the same units. The share of instances certified at the root box with no subdivision at all rises from 34.8% to 52.8%. Figure 6 shows the paired comparison.

**Fixed-budget certification (E2).** On the locked test split (72 instances), mean $\rho^{\mathrm{lo}}$ is 0.0000 for A0, 0.0330 for A1, 0.0534 for A2 and 0.0556 for A3. The share of instances with a strictly positive certified radius moves from 0.0% to 94.4%. The prespecified contrasts are

$$\mathrm{P1} = 0.0330\ [0.0233,\ 0.0437], \qquad \mathrm{P2} = 0.0204\ [0.0150,\ 0.0264],$$

with 95% paired bootstrap intervals over instances, win/loss/tie counts 67/0/5 and 68/0/4, probabilities of superiority 0.965 and 0.972, and Holm-adjusted $p$-values 1.52e-12 and 1.52e-12. Decision-relevant branching contributes a further 0.0023 [0.0001, 0.0043], reported as exploratory. On the locked stress split the corresponding means are 0.0000 and 0.0555. Figure 7 shows the distributions.

**Anytime behaviour (E7) and brackets.** Mean certified volume of $\mathcal{B}(0.02)$ after 64, 256 and 1,024 processed boxes is 0.000, 0.000 and 0.000 for A0 against 0.529, 0.557 and 0.589 for A3. A verified switch witness was found for 136 of 192 instances; among those the median bracket width is 0.083 in $p$, and the number of instances whose certified lower bound exceeded its own witness – which would be a contradiction – is 0. Figure 8 shows both.

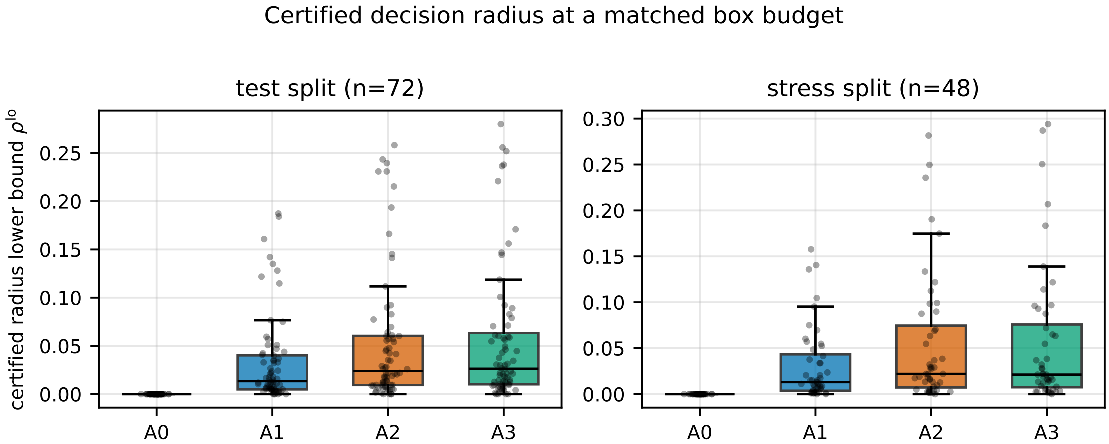


**Figure 7:** Certified lower bound on the decision radius at a matched budget of 300 processed boxes, on the two locked splits. A0 is the all-candidate retention condition, A1 adds decision focus, A2 adds shared-anchor enclosures, A3 adds decision-relevant branching. Boxes show quartiles with the median; points are individual instances. A value of zero means "not certified within this budget", not a radius of zero.

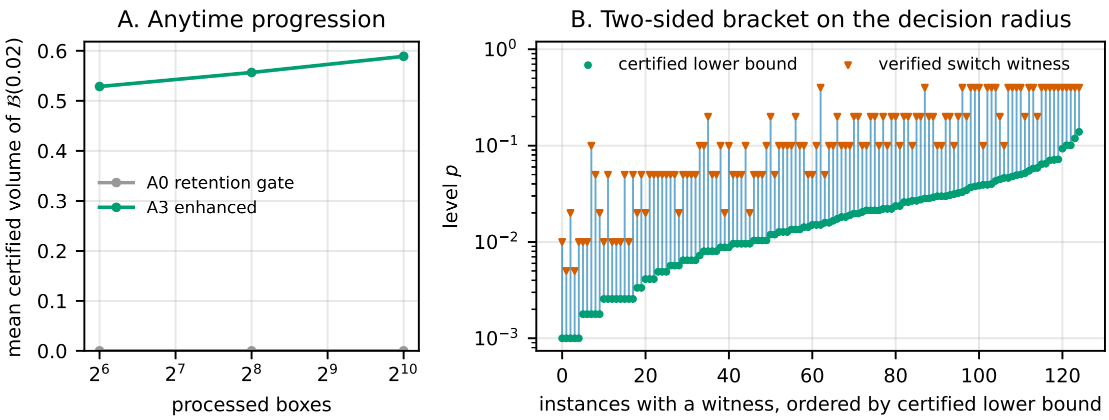


**Figure 8:** Anytime behaviour and the two-sided radius bracket. Panel A: mean certified volume of the focal box against the processed-box budget, for the all-candidate retention condition and for the enhanced engine; certified volume is nondecreasing in the budget because certified leaves are never revisited. Panel B: for each instance with a verified switch witness and a strictly positive certified radius, the interval between the certified lower bound and the smallest level at which the exact rational oracle names a different unique winner. A bar's height is what remains unresolved between proof and counterexample, not an error bar.

**What this does not show.** These are certification results and nothing more. They do not show that ReLexTail chooses better than another method, that its recommendations carry lower external regret, or that any calibration guarantee holds; the corresponding tracks were not run, and Section 8 records them as unestablished. The instances are generated, the shortlist size is declared rather than elicited, and the stress families test extrapolation without licensing any claim about arbitrary deployment shift. A radius reported as zero always means "not certified within this budget".

# 7 Computational study and supplier illustration

## 7.1 A matched, reproducible protocol

The study isolates the behaviour of the finite selection rule. It comprises 90 synthetic candidate sets: three front geometries, three criterion counts ($m = 3, 6, 10$), and ten independent replicates per combination. Each set contains 80 candidates. Linear fronts use $f = 0.5s$ for a uniformly Dirichlet-distributed simplex vector $s$. Convex fronts use $f_i = s_i^2$. Concave fronts normalise a vector of square-root-transformed independent uniform draws to Euclidean norm one. These are controlled synthetic geometries, not exact solutions of named engineering problems or real procurement datasets. Every generated matrix is saved.

The nominal criterion bounds and probe anchors are computed from the candidate set. The family consists only of singletons, the mean and the maximum, subject to the stated retention test. Retention is checked to remain identical on every sampled perturbation. Eight common resolution widths are tested: $0.001, 0.005, 0.01, 0.02, 0.03, 0.05, 0.075, 0.1$. The comparators are exact LexPR on $D$; exact leximax and Chebyshev selection on the normalised criterion vector; Mean-D; upper-25% and upper-50% tail minimisation on $D$; and a uniform weighted sum on the normalised criteria. They separate changes in order, tail aggregation and representation. Chebyshev and leximax happen to select identically throughout the saved population, so only leximax is shown in the compact table. All 15 method variants remain in the data. These are controlled comparators, not an exhaustive ranking of MCDA methods.

At each level $p \in \{0.01, 0.025, 0.05, 0.10, 0.15, 0.20, 0.30, 0.40\}$, 500 uniform bound perturbations are shared across all methods. The same random directions are scaled across levels. This yields 45,000 matched selections per method and level-specific summaries for every instance. Within an instance, one common set of 10,000 Dirichlet($\mathbf{1}$) weight vectors evaluates every candidate on the fixed nominal criterion scale. The master seed is 20260905, and instance streams are indexed by geometry, criterion count and replicate. Deterministic ties use the smallest row identifier.

For a test weight vector $w$, the linear loss and its range-normalised regret are

$$L_w(x) = \sum_i w_i r_i(x; b_0), \qquad \Delta L_w(x) = \frac{L_w(x) - \min_{y \in A} L_w(y)}{\max_{y \in A} L_w(y) - \min_{y \in A} L_w(y)}. \tag{13}$$

If the denominator is zero, regret is defined as zero for all candidates for that weight. Mean regret averages all 10,000 values. Upper-tail regret averages exactly the 2,500 largest values. The latter is empirical upper-25% CVaR, including the appropriate mass when values tie. Candidate losses are computed once on the nominal scale. We then evaluate the points selected under perturbed bounds and average first over draws within each candidate set, then equally over candidate sets.

Point-change rate is the fraction of draws changing the fully refined point relative to its nominal identifier. Category-set change rate compares the complete nominal and perturbed $W_{\text{cat}}$ sets. We also save category-set size, Jaccard overlap and retention of the nominal point. None of these set metrics is substituted for a point-change rate in cross-method comparisons. Nominal divergence compares the nominal ReLexTail and LexPR identifiers.

Table 4 and Figure 9 use the same 90-instance population at $p = 0.05$. Confidence intervals resample paired instance summaries, preserving each instance's methods and all 500 perturbations as one block. Table intervals use 10,000 percentile-bootstrap replicates. For two focal outcomes at $\delta = 0.02$, Table 5 uses 20,000 paired bootstrap replicates, a paired Wilcoxon signed-rank check, probability of superiority with half credit for ties, and Holm correction across exactly those two checks. Seed 20260906 fixes both resampling analyses. The focal width represents two percentage points of active range and was retained from the earlier protocol; the analysis was not preregistered, so inferential quantities remain descriptive. No resolution is declared universally optimal.

## 7.2 Results and limits of the trade-off

At $\delta = 0.02$, ReLexTail changes its point on 19.30% of draws compared with 20.88% for LexPR. The paired mean difference is $-1.584$ percentage points, but its 95% bootstrap interval $[-3.482, 0.091]$ includes zero and the Holm-adjusted Wilcoxon $p$-value is 0.148. Thus the enlarged run does not support a clear point-stability improvement. Mean external upper-tail regret is 0.5411 versus 0.5428; the paired difference is $-0.00177$ with interval $[-0.00329, -0.00065]$, superiority probability 0.722 and adjusted $p = 4.44 \times 10^{-7}$. This effect is consistent on this population but small in magnitude and conditional on the held-out weight distribution.

Exact leximax obtains lower tail regret (0.5340) but a higher point-change rate (35.64%). Tail25-D attains tail regret 0.5401 and a 20.84% point-change rate, showing that most of the numerical tail benefit does not require categories. The uniform weighted sum attains the lowest mean regret

**Table 4:** Paired comparison on 90 candidate sets at $p = 0.05$. Regret is evaluated on the nominal scale after perturbed selection. The interval is a 95% paired instance-bootstrap interval for the tail-regret difference from LexPR. Point changes are percentages.

| Method | Mean regret | Tail regret | Difference interval | Point changes |
|---|---|---|---|---|
| Weighted-sum | 0.2915 | 0.6270 | [+0.05677, +0.11376] | 19.52 |
| Mean-D | 0.3070 | 0.5491 | [+0.00009, +0.01275] | 30.50 |
| Leximax | 0.4136 | 0.5340 | [-0.01216, -0.00556] | 35.64 |
| Tail25-D | 0.3800 | 0.5401 | [-0.00473, -0.00097] | 20.84 |
| LexPR | 0.3810 | 0.5428 | [+0.00000, +0.00000] | 20.88 |
| ReLexTail $\delta = 0.005$ | 0.3810 | 0.5425 | [-0.00092, +0.00011] | 19.96 |
| ReLexTail $\delta = 0.01$ | 0.3810 | 0.5419 | [-0.00181, -0.00025] | 20.16 |
| ReLexTail $\delta = 0.02$ | 0.3799 | 0.5411 | [-0.00324, -0.00064] | 19.30 |
| ReLexTail $\delta = 0.05$ | 0.3741 | 0.5407 | [-0.00362, -0.00093] | 21.21 |
| ReLexTail $\delta = 0.075$ | 0.3729 | 0.5401 | [-0.00422, -0.00129] | 20.78 |
| ReLexTail $\delta = 0.1$ | 0.3731 | 0.5407 | [-0.00401, -0.00054] | 23.24 |

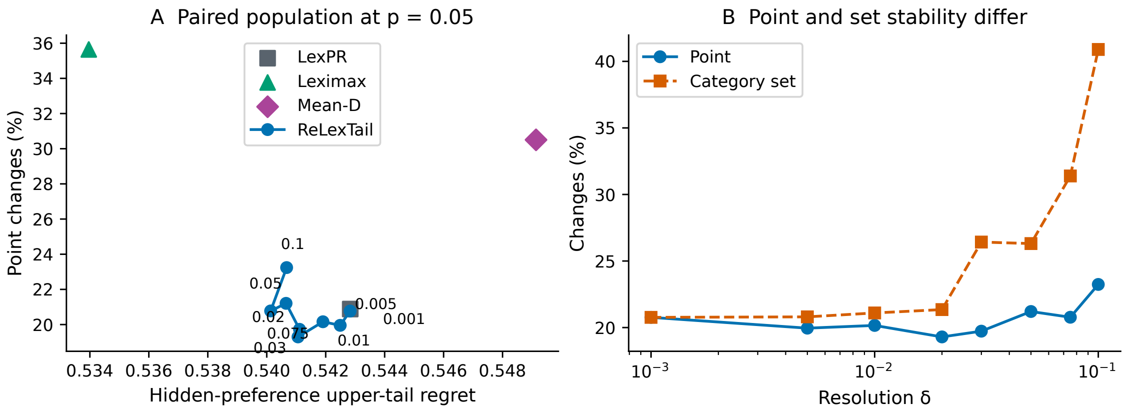


**Figure 9:** Matched quality–stability results at $p = 0.05$ on all 90 synthetic instances. Lower values are preferred on both axes in panel A; labels on the ReLexTail curve give resolution widths. Panel B separates changes to the point from changes to the category-optimal set. Connecting lines identify tested resolutions and do not imply monotone behaviour between them.

**Table 5:** Two focal paired effects for ReLexTail at $\delta = 0.02$ minus LexPR ($p = 0.05$, 90 instance summaries). Probability of superiority counts a lower ReLexTail value as a win and gives half credit to ties. W/L/T are wins/losses/ties. Holm correction covers the two displayed Wilcoxon checks.

| Outcome | Mean difference | 95% bootstrap interval | Superiority | W/L/T | Holm $p$ |
|---|---|---|---|---|---|
| Point changes (pp) | -1.584 | [-3.482, +0.091] | 0.572 | 32/19/39 | 0.148 |
| Upper-tail regret | -0.002 | [-0.003, -0.001] | 0.722 | 48/8/34 | $4.44 \times 10^{-7}$ |

(0.2915) and a much worse upper-tail regret (0.6270), illustrating that average performance and protection across hidden preferences are different objectives.

Coarser resolution is not uniformly more stable. At $\delta = 0.1$, ReLexTail's point-change rate is 23.24%, above the LexPR rate, even though its mean tail regret is slightly lower. Category-set change rises to 40.90%, with a mean perturbed category-set size of 2.90. At $\delta = 0.02$, category-set change is 21.35% and mean size is 1.08. A larger set is therefore not automatically a more stable set under exact set equality. The category-stability theorem supplies a conditional local guarantee, not an empirical promise that these sets never change.

Figure 10 disaggregates the main comparison rather than assuming the pooled effect holds for every geometry. Figure 11 separately reports nominal disagreement. A method can share a nominal winner with another method yet respond differently to perturbed bounds. Conversely, disagreement alone does not establish better quality. These distinctions prevent nominal selection, external regret

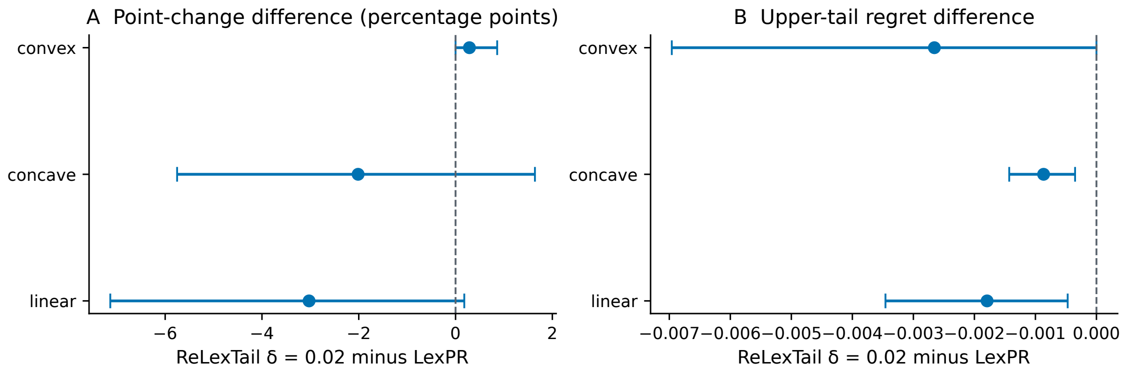


**Figure 10:** Family-specific paired differences between ReLexTail at $\delta = 0.02$ and LexPR, at $p = 0.05$. Each family contains 30 candidate sets spanning three criterion counts. Points are mean differences; bars are 95% percentile-bootstrap intervals from 4,000 paired instance resamples. Negative values favour ReLexTail. The intervals are descriptive and not adjusted for multiple comparisons.

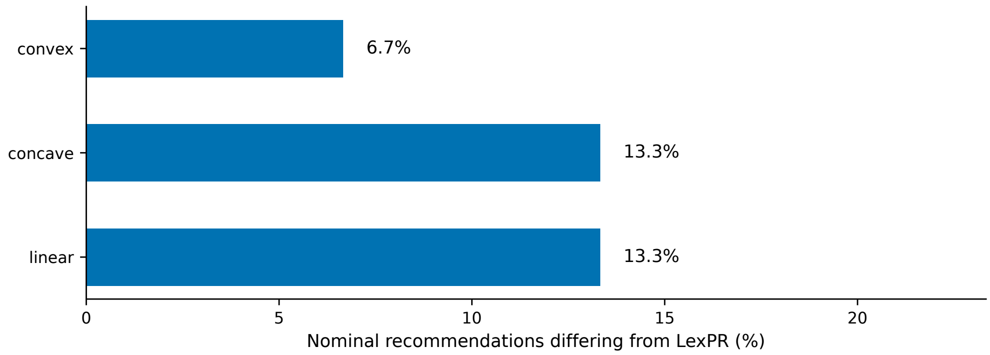


**Figure 11:** Nominal recommendation disagreement between ReLexTail at $\delta = 0.02$ and exact LexPR, by synthetic geometry. Each denominator is 30 candidate sets. This concerns unperturbed recommendations and is distinct from the point-change rates under perturbed bounds.

and robustness from being treated as interchangeable outcomes.

## 7.3 Grid, representation and public-data checks

Figure 12 tests three potential alternative explanations. Across four grid origins, point changes for $\delta = 0.02$ range from 19.30% to 20.65%; the result is therefore sensitive in magnitude but not created by the unshifted origin. Using only $C_M$, then $C_M, C_{25}$, or all four category coordinates gives similar point-change rates (19.40%, 19.40% and 19.30%) while category-set changes fall from 28.54% to 21.35%. Removing all singleton probes changes the representation and raises point changes to 37.22%; it is not a neutral simplification. Against each nominal winner's nearest full-profile rival, the first difference occurs at $C_M$ on 64 instances, $C_{25}$ on 13, $C_{50}$ on six and exact $T_{25}$ on seven. No later coordinate is decisive in this population.

For an application-oriented reproducibility check, we downloaded the UCI Energy Efficiency dataset (Tsanas & Xifara, 2012). It contains 768 simulated building designs, not measurements from occupied buildings. We minimise surface area, heating load and cooling load; surface is only a material-use proxy, not a monetised or environmental impact. Removing duplicate objective profiles and Pareto-dominated rows leaves four unique candidates. On the unperturbed front, LexPR, Mean-D, the uniform weighted sum and every tested ReLexTail width select saved source row 26. At $p = 0.05$, LexPR changes on 32.6% of 500 draws and ReLexTail at $\delta = 0.02$ on 32.0%; at $\delta = 0.05$ ReLexTail happens not to change, whereas neighbouring widths do. This isolated discontinuity demonstrates boundary dependence, not superior real-world performance. The archived source, licence, checksum, filtering and all perturbed results are included.

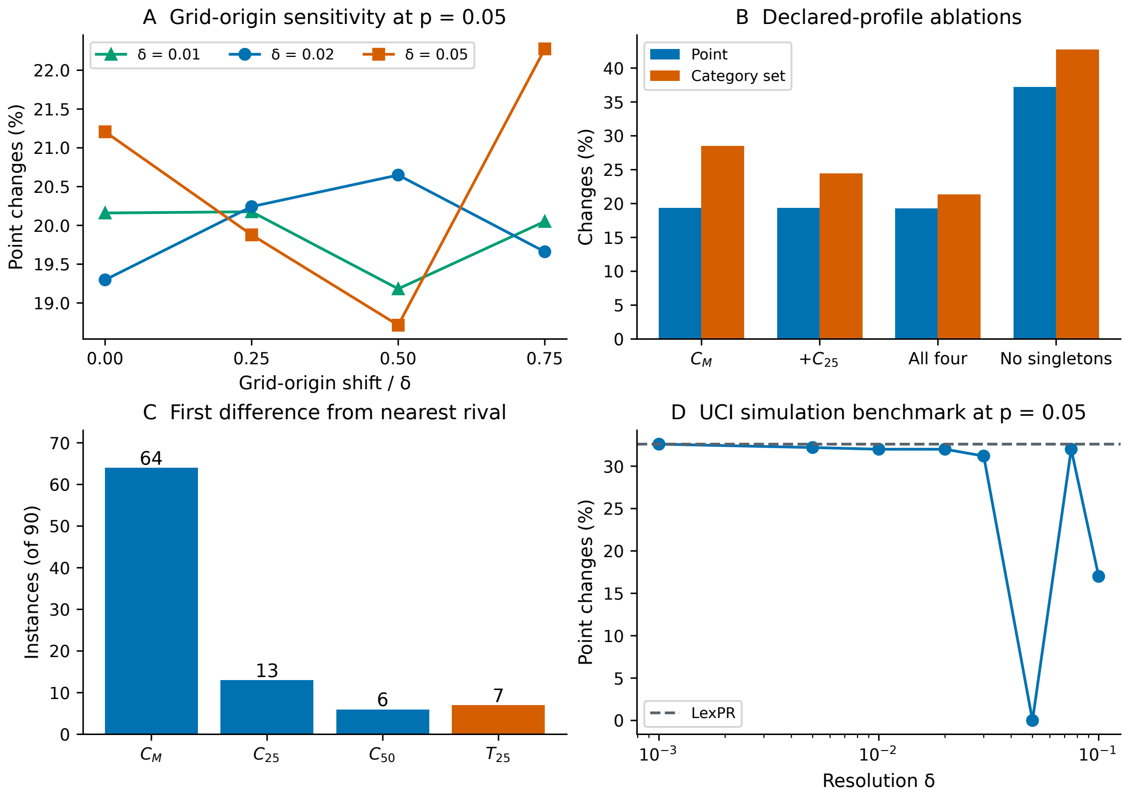


**Figure 12:** Additional checks using the frozen synthetic directions and a public simulation dataset. Panel A shifts fixed-cell origins at three resolutions. Panel B changes the category prefix or removes singleton probes. Panel C identifies the first full-profile difference from the nearest nominal rival. Panel D is a single UCI building-simulation instance and must not be read as population evidence; its nonmonotone jump illustrates why widths require sensitivity analysis.

## 7.4 A constructed supplier decision

The supplier illustration contains ten named alternatives and seven criteria to minimise: unit cost, defect rate, lead time, greenhouse-gas emissions, energy intensity, social risk and supply-resilience risk. Values are fixed in the accompanying source and are constructed, not field measurements. The instance illustrates how a decision analyst can expose the decisive concern and investigate sensitivity. It does not validate the model's preferences for a procurement organisation.

With nine retained canonical probes and common resolution $\delta = 0.02$, ReLexTail selects S7. Figure 13 compares its labelled disappointments with S6, the nearest rival under the full profile. Both have maximum disappointment 0.787879, hence maximum category 40. Their upper-25% tail means are 0.715814 and 0.743146 respectively, giving categories 36 and 38. Thus the decisive coordinate against S6 is $C_{25}$, not $C_M$. S1 has maximum category 41 and is eliminated earlier. Reporting only the S7–S1 contrast would miss the closest rival and misrepresent the actual hierarchy.

The distance $0.800000 - 0.787879 = 0.012121$ is only S7's upward distance to its maximum-category boundary. It is not the global $\mu_{\text{cell}}$ of the stability theorem. On this instance the global minimum distance is zero because some risk coordinates lie exactly on boundaries. The theorem's sufficient condition is therefore uninformative without a more refined treatment of structurally invariant coordinates. A positive bound-robustness guarantee must come from a direct certificate rather than that single distance.

At $p = 0.05$, three of the 300 draws change the nominal supplier. At $p = 0.20$, the rate is 30.67%. S1 and S7 are the only sampled winners from $p = 0.05$ through $p = 0.50$. This does not prove that no other supplier can win in those boxes. Sampling and the interval illustration in Section 5 use the same type of range-relative perturbation, but they concern different candidate matrices and must not be combined into a single empirical certificate.

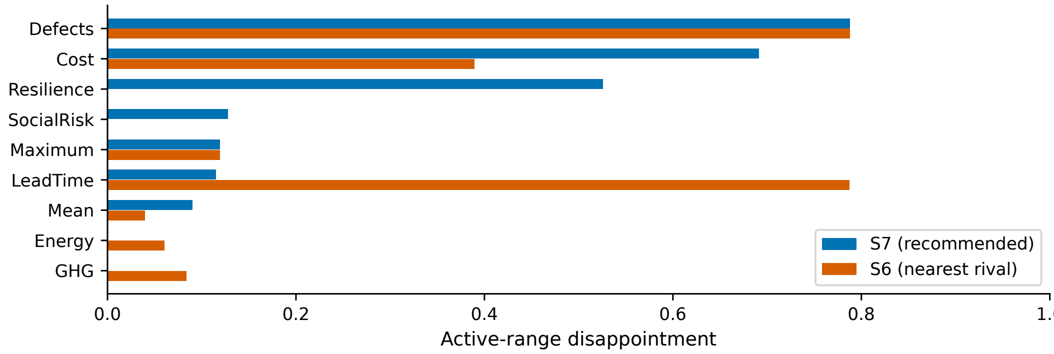


**Figure 13:** Labelled probe disappointments for the recommended supplier S7 and its nearest rival S6. Values are recomputed from the saved criterion matrix with nine canonical probes. Bars explain the individual concerns; optimality follows from comparing complete profiles across all ten candidates.

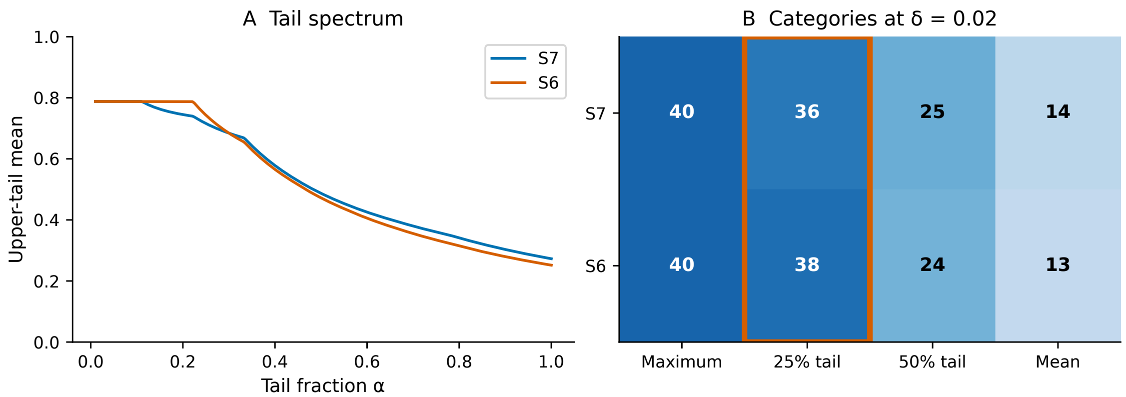


**Figure 14:** Supplier tail spectrum and category comparison at $\delta = 0.02$. S7 and S6 tie in maximum category, but S7 has a lower upper-25% tail category, highlighted in panel B. Later coordinates need not favour S7 once that first strict difference settles the comparison.

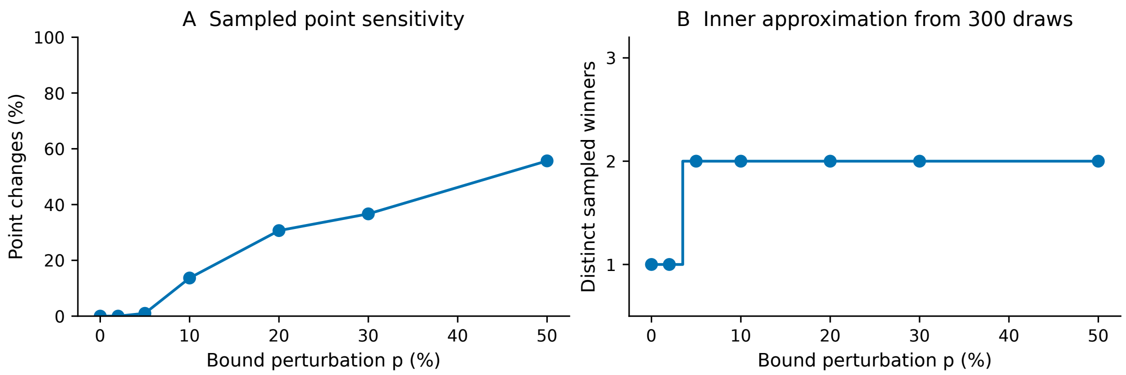


**Figure 15:** Supplier sampling sensitivity for ReLexTail at $\delta = 0.02$, using 300 independent uniform bound draws at each level and nominal-range perturbations. Panel A reports point changes. Panel B reports distinct observed winners, an inner sampling approximation rather than a certified outer bound. The $p = 0.5$ endpoint is sampled only and is excluded from closed-box certification.

# 8 Discussion and conclusions

ReLexTail offers a specific form of noncompensatory decision aiding: a deterioration to a worse maximum category cannot be offset by improvements elsewhere, while differences inside that category may be deferred in favour of broader-tail concerns. This is a preference commitment. An analyst should elicit whether that commitment is appropriate before selecting the probe family or widths. The method is most useful when named concerns, finite resolution and a reproducible explanation are important parts of the decision process.

Resolution is not a statistical confidence level or a solver tolerance. It sets category widths on the active-range disappointment scale. A width of 0.02 means two percentage points of a probe's available range, not two percentage points of a physical criterion. Equal widths across risk coordinates are convenient for comparison but need not fit every application. A practical report should record the elicitation exercise, state the chosen widths, compare nearby widths and explain whether category membership or the final point is the intended recommendation. The four-origin analysis shows that the focal point-change rate remains within a narrow two-percentage-point band, but other widths and the UCI example remain visibly boundary-dependent. Grid sensitivity is therefore a required diagnostic, not a proof of robustness.

The finite-set properties are conditional on the declared representation. Positional anonymity does not make the rule preference-free, since repeated probes change their influence. Pareto compatibility requires retained separating probes. Active anchors introduce dependence on the candidate set. Category stability concerns perturbations that avoid every relevant boundary and does not imply stability of exact refinements. The method's auditability comes from exposing these conditions and replaying its comparisons, not from eliminating modelling choices.

The computational study shows modest tail-regret gains over exact LexPR at some widths and deterioration in point stability at a coarser width. The enlarged paired analysis does not establish a point-stability gain at $\delta = 0.02$, because its interval includes zero. Tail25-D also captures much of the external tail-regret difference, so the categories should be justified by resolution semantics and set-valued reporting rather than by that metric alone. These findings argue for reporting the full trade-off rather than selecting a favourable metric. The three synthetic geometries and uniform linear test preferences restrict external validity. The public building data are themselves simulated and reduce to four nondominated objective profiles under our chosen objectives; they provide reproducibility and a boundary stress test, not stakeholder or field validation. Operational applications, additional utility families and prospective elicitation studies would be needed to establish practical benefit.

Interval analysis addresses a different uncertainty question. Its sound outer set can remain large even when random draws repeatedly select one point. This is an informative distinction between absence of observed switches and proof that switching is impossible. Section 6 narrows that gap along the two axes where the earlier engine was conservative for reasons unrelated to the decision. Requiring all candidates to retain all categories is sufficient but not necessary, and Proposition 5 exhibits an instance where it certifies nothing while the recommendation is certain over the whole box; widening the shared normalisation anchors independently is sound but loose, and Proposition 7 exhibits an instance where that slack alone decides whether the box is certified. Neither change touches the declared rule, and neither is a new result in interval analysis: what is new is the identification of which shared quantities occur in this rule and the measured consequence for certification. The continuous-polyhedral formulation establishes a computational characterisation, while independent numerical validation of a full solver implementation remains future work.

The scope of the certification evidence should not be overstated. It concerns generated instances on a declared shortlist size, with a resolution and a focal level fixed in advance; a certified radius reported as zero means only that the budget did not suffice. Nothing in it bears on decision quality. Whether the recommendation carries lower external regret than a competing method at matched coverage, and whether a calibrated emission policy can meet a prespecified risk target, are separate statistical claims that require a separate experimental track with independent decision episodes; that track was not run here, and no part of the certification result substitutes for it. The upper-tail regret comparison reported in Section 7 remains a small, metric-dependent effect on one synthetic population. Table 6 states, for each claim the paper makes, what kind of evidence supports it.

The paper establishes a complete and transitive resolution-aware preorder, conditional Pareto

**Table 6:** Claim-to-evidence register. "Proved" means a proof appears in this paper; "measured" means the number is produced by the released scripts from released data; "unestablished" means the paper does not claim it.

| Claim | Status | Evidence |
|---|---|---|
| ReLexTail is a complete transitive preorder with exact refinement | proved | Section 4 |
| Conditional Pareto compatibility and replication invariance | proved, conditional | Section 4, stated assumptions |
| Category stability under a sup-norm perturbation | proved, sufficient only | Theorem 5 |
| Interval elimination is sound | proved | Proposition 2 |
| A decision can be certified with no category retained | proved | Proposition 4 |
| The all-candidate condition can certify nothing while the decision is certain | proved by exhibition | Proposition 5, rational example |
| Shared-anchor enclosures are contained in independent ones | proved | Proposition 6 |
| That containment can decide certification | proved by exhibition | Proposition 7, rational example |
| No emitted certificate was falsified | measured | Section 6.5, exact oracle and independent replay |
| Decision focus and dependency preservation raise the certified radius | measured, locked split | Section 6.5, paired contrasts P1 and P2 |
| Upper-tail regret differs from exact LexPR at $\delta = 0.02$ | measured, small | Section 7, paired interval |
| Point-stability gain at $\delta = 0.02$ | not supported | paired interval includes zero |
| Sequential polyhedral characterisation | proved | Section 5 |
| End-to-end continuous solver | unestablished | no validated implementation |
| Lower external regret at matched coverage | unestablished | track not run |
| Calibrated risk or coverage guarantee | unestablished | track not run |
| Benefit on measured operational data | unestablished | no field data with repeated observations |
| Priority over the closest literature | unestablished | full-text verification pending |

compatibility, replication invariance, a local category-stability guarantee, and a decision-focused dependency-preserving certificate with a two-sided radius bracket and an independent replay path. Its numerical evidence identifies a qualified, metric-dependent quality trade-off on the stated synthetic population rather than a general stability advantage, together with a clearer certification result on a separate locked benchmark. The supplier illustration makes the decisive upper-tail comparison explicit. Together these results support ReLexTail as a transparent modelling option when the resolution of declared concerns matters, with sensitivity, unresolved uncertainty and the untested claims reported alongside the recommendation.

# Declarations

**Author contributions (CRediT).** Madani Bezoui: conceptualization, methodology, software, formal analysis, investigation, visualization, writing–original draft, and writing–review and editing.
**Funding and acknowledgments.** This research received no external funding. No additional acknowledgments are declared.
**Competing interests.** The author declares no competing interests.
**Ethics.** No human participants, animals or personal data were involved. The supplier data are constructed.
**Data and code availability.** The submission source package includes the scripts, generated candidate matrices, perturbation directions, protocol, per-instance results, statistical summaries, figure sources and the UCI Energy Efficiency file under CC BY 4.0 with its SHA-256 checksum. The UCI dataset is independently available at `https://doi.org/10.24432/C51307`. The

complete code, data and reproduction scripts are permanently archived under the concept DOI `https://doi.org/10.5281/zenodo.21771785` (which always resolves to the latest version), and the active project repository is `https://github.com/MadBezoui/ReLexTail`. It supersedes the earlier and smaller release archived at `https://doi.org/10.5281/zenodo.21771786`, to which the experiments reported here must not be attributed. The complete local reproduction command is documented in the package README.

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